TOOLS.MAHFUJ.BD

Project Documentation

Website hosting and deployment overview

Prepared for the administration and future maintenance of the tools.mahfuj.bd website.

Project environment: Rocky Linux · Apache HTTP Server · Nginx Proxy Manager · Cloudflare

1. What this project does

This project makes the tools website available at https://tools.mahfuj.bd. The website files live on the existing Rocky Linux server. Apache serves those files on the private network, while Nginx Proxy Manager (NPM) receives public web requests, handles HTTPS and passes the requests to Apache. Cloudflare manages the domain's DNS.

The arrangement keeps the web server behind the reverse proxy. Visitors use the public domain; they do not need to connect directly to the Rocky Linux server.

2. Current project details

Item	Current value
Website address	tools.mahfuj.bd
Rocky Linux server name	host.mahfuj.bd
Rocky Linux private address	192.168.50.*/24
Web server	Apache (httpd)
Website directory	/var/www/tools.mahfuj.bd/html
Apache site configuration	/etc/httpd/conf.d/tools.mahfuj.bd.conf
Reverse proxy	Nginx Proxy Manager
Public certificate	Let's Encrypt, managed in NPM
DNS provider	Cloudflare

3. How a visitor reaches the website

A normal request follows this path:

```
Visitor's browser
      |
      | HTTPS request for tools.mahfuj.bd
      v
Cloudflare DNS / proxy (if proxying is enabled)
      |
      v
Public IP and edge router
      |  TCP 80 and 443 forwarded to NPM
      v
Nginx Proxy Manager
      |  HTTP to 192.168.50.*:80
      v
```

```
Rocky Linux / Apache
      |
      v
/var/www/tools.mahfuj.bd/html
```

NPM is the public-facing reverse proxy in this design. It selects the correct Proxy Host using the requested domain, presents the TLS certificate and forwards the request to Apache over the LAN.

Hostname note: host.mahfuj.bd is the Rocky Linux machine's hostname. tools.mahfuj.bd is the website's hostname in Apache. They are separate settings and do not need to match.

4. Apache configuration

Apache uses a name-based VirtualHost for tools.mahfuj.bd. The site configuration is stored in /etc/httpd/conf.d/tools.mahfuj.bd.conf and points to the website's html directory.

```
<VirtualHost *:80>
    ServerName tools.mahfuj.bd
    DocumentRoot /var/www/tools.mahfuj.bd/html

    <Directory /var/www/tools.mahfuj.bd/html>
        Options -Indexes +FollowSymLinks
        AllowOverride All
        Require all granted
    </Directory>

    ErrorLog /var/log/httpd/tools-error.log
    CustomLog /var/log/httpd/tools-access.log combined
</VirtualHost>
```

Apache serves the origin over HTTP on the private network. Public HTTPS is handled by NPM. If the design changes and Apache is exposed directly, review the TLS and firewall configuration first.

5. Website files

The expected directory layout is:

```
/var/www/tools.mahfuj.bd/html/
├── index.html
├── css/
│   └── style.css
├── js/
│   └── script.js
└── images/
    └── (image assets)
```

The names above are examples of the expected layout. Keep the paths in index.html consistent with the files actually uploaded. Linux filenames are case-sensitive. Root-relative links such as /css/style.css and /js/script.js are suitable when the site is served from the domain root.

6. DNS, reverse proxy and HTTPS

Cloudflare should have an A record named tools pointing to the current public IPv4 address. The router should forward inbound TCP ports 80 and 443 to the NPM host. Do not forward those public ports to the Rocky Linux origin when NPM is intended to handle public traffic.

NPM Proxy Host setting	Value
Domain	tools.mahfuj.bd

Forward scheme	http
Forward host / IP	192.168.50.*/24
Forward port	80
SSL certificate	Let's Encrypt certificate for tools.mahfuj.bd

After the certificate has been issued in NPM, enable Force SSL and HTTP/2 Support. If certificate validation fails, check DNS, router forwarding and the selected validation method. A DNS challenge may be more suitable when Cloudflare proxying prevents HTTP-01 validation.

7. Security and maintenance

- Keep Rocky Linux and Apache updated using the supported Rocky Linux repositories.
- Keep public HTTP and HTTPS traffic directed through NPM. Avoid exposing Apache or the NPM administration interface unnecessarily.
- Leave SELinux enabled. Restore the normal web-content labels after copying website files.
- Use sensible file permissions; the web server should not have write access to the whole project or system.
- Keep a backup of the website files, Apache VirtualHost file and NPM configuration/certificate data.
- Review /var/log/httpd/tools-access.log and /var/log/httpd/tools-error.log when diagnosing site problems.

8. Checked before handing over

- Cloudflare resolves tools.mahfuj.bd to the intended public endpoint.
- The router forwards TCP 80 and 443 to the NPM host.
- NPM forwards tools.mahfuj.bd to 192.168.50.* on port 80.
- Apache is enabled, running and passes apachectl configtest.
- The site opens over HTTPS and the browser trusts its certificate.
- CSS, JavaScript and image files load correctly.
- The services return after a server restart and backups are available.

Result: https://tools.mahfuj.bd

(**Note**: if you don't find the site using this link, probably I have deleted the machine)

